IT Project Management

Topic 3: Project Planning & Controlling

Diploma Level | Module 3.1 - 3.5

3.1 Project Objectives

Objectives define what needs to be achieved. In IT projects, we adhere to the **SMART** criteria to ensure clarity.

Specific

Clear & unambiguous goals.

Measurable

Quantifiable progress.

Achievable

Realistic within resources.

Time-Bound

Defined deadlines.

3.2 Work Breakdown Structure (WBS)

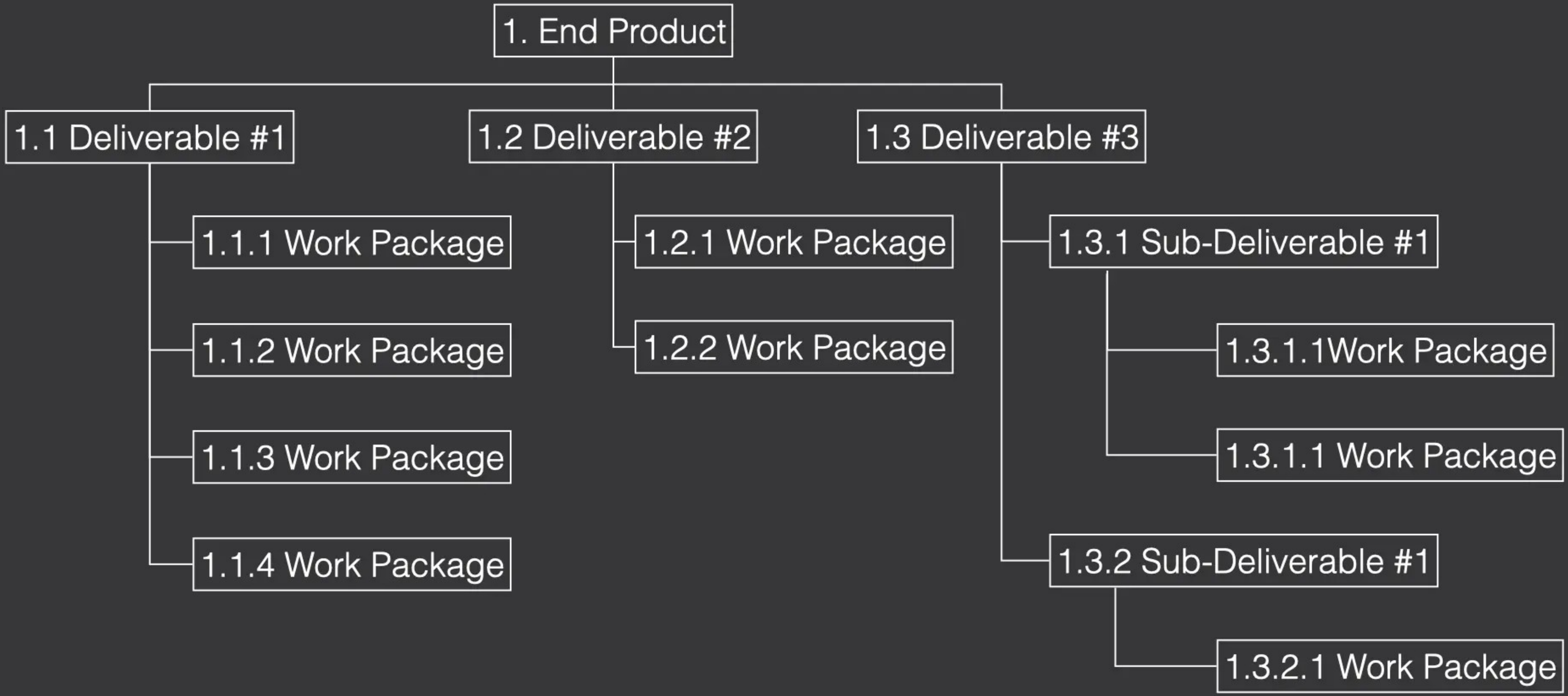
Decomposition of Work

The WBS is a hierarchical decomposition of the total scope of work to be carried out by the project team.

- **Root:** The final project deliverable (e.g., "Mobile App").
- **Phases:** Major logical groupings (Design, Dev, Test).
- **Work Packages:** The lowest level, manageable activities.

The 100% Rule: The WBS includes 100% of the work defined by the project scope.

Work Breakdown Structure Chart



3.3 Responsibility Matrix (RACI)

The RACI matrix clarifies roles and responsibilities for each task, preventing confusion in IT teams.

Activity	Project Manager	Developer	Tester	Client
Define Requirements	Accountable	Consulted	Informed	Responsible
Code Module A	Accountable	Responsible	Informed	Informed
System Testing	Informed	Consulted	Responsible	Informed
User Acceptance	Accountable	Consulted	Consulted	Responsible

R: Responsible (Doer), **A**: Accountable (Owner), **C**: Consulted (Two-way), **I**: Informed (One-way)

3.4 Scheduling & 3.4.1 Duration Estimates

Project Scheduling

Scheduling determines the start and finish dates for project activities. It is the roadmap for execution.

- Identify activities (from WBS).
- Sequence activities (Dependencies).
- Estimate resources & durations.

Estimation Techniques

- **Expert Judgment:** Based on historical experience.
- **Analogous:** Comparing to a similar past project (Top-down).
- **Parametric:** Statistical relationship (e.g., hours per line of code).
- **Three-Point:** (Optimistic + 4*MostLikely + Pessimistic) / 6.

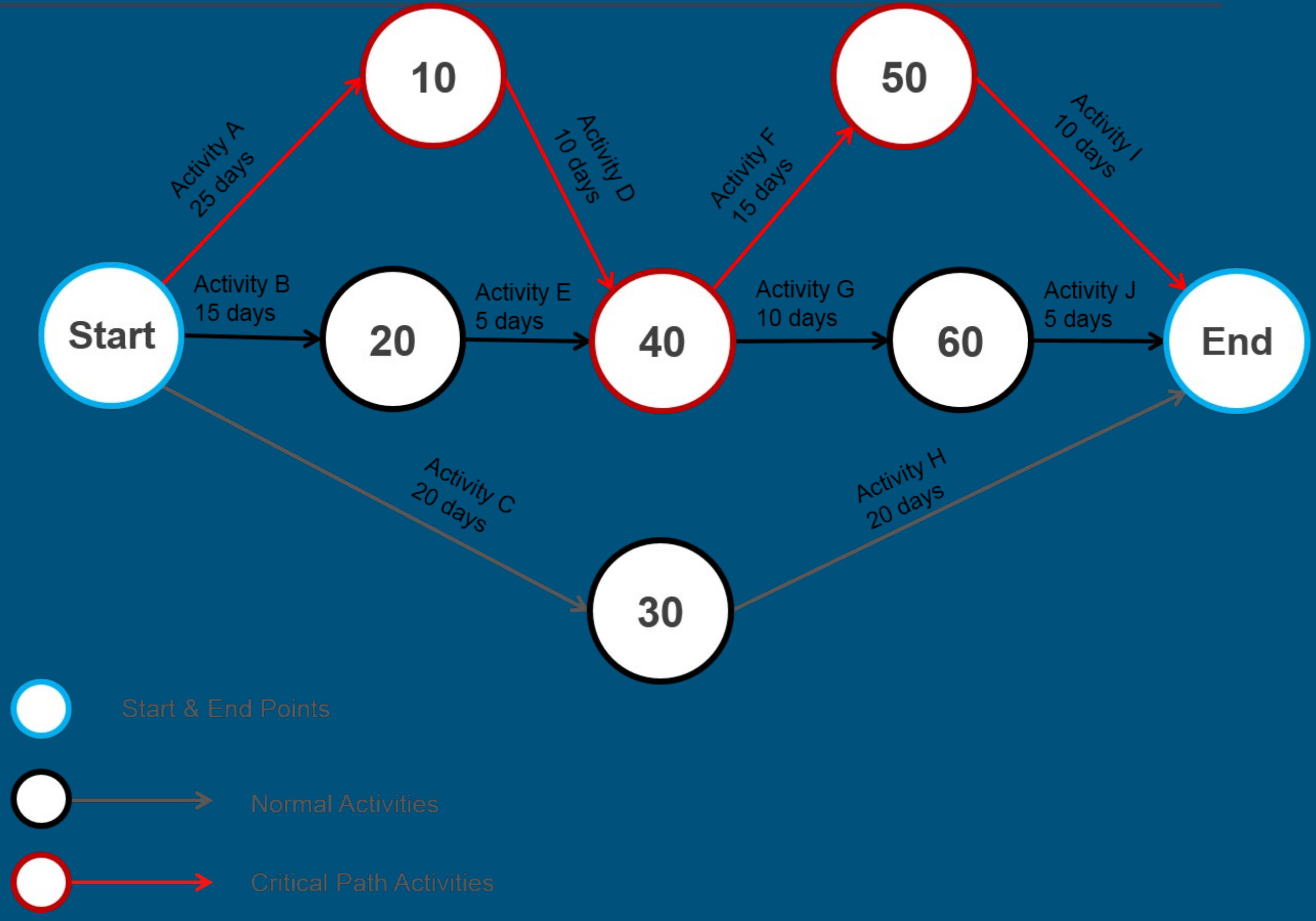
3.5 Developing the Network

Network Logic

A network diagram visually displays the logical relationships (dependencies) between project activities.

- **Nodes:** Represent activities (in PDM - Precedence Diagramming Method).
- **Arrows:** Show dependencies (A must finish before B starts).
- **Path:** A sequence of connected activities.

This allows us to identify the critical path and calculate project duration.



3.4.3 Forward Pass Calculation

Concepts

The Forward Pass determines the earliest times an activity can start and finish based on network logic.

- **ES (Early Start):** The earliest point an activity can begin.
- **EF (Early Finish):** ES + Duration.
- **Rule:** When activities merge, the ES of the successor is the **MAX** EF of all predecessors.

Formulas

$$EF = ES + \text{Duration}$$

$$ES_{\text{successor}} = \max\left(EF_{\text{predecessors}}\right)$$

3.4.3 Backward Pass Calculation

Concepts

The Backward Pass determines the latest times an activity can start and finish without delaying the project.

- **LF (Late Finish):** The latest point an activity can end.
- **LS (Late Start):** LF - Duration.
- **Rule:** When tracing back, the LF of a predecessor is the **MIN** LS of all successors.

Formulas

$$LS = LF - Duration$$

$$LF_{predecessor} = \min\left(LS_{successors}\right)$$

Critical Path & Slack

0

Total Slack

The amount of time an activity can be delayed without delaying the project end date.

Slack = LS - ES

CP

Critical Path

The longest path through the network diagram. It determines the shortest time to complete the project. Activities on the Critical Path have **Zero Slack**.

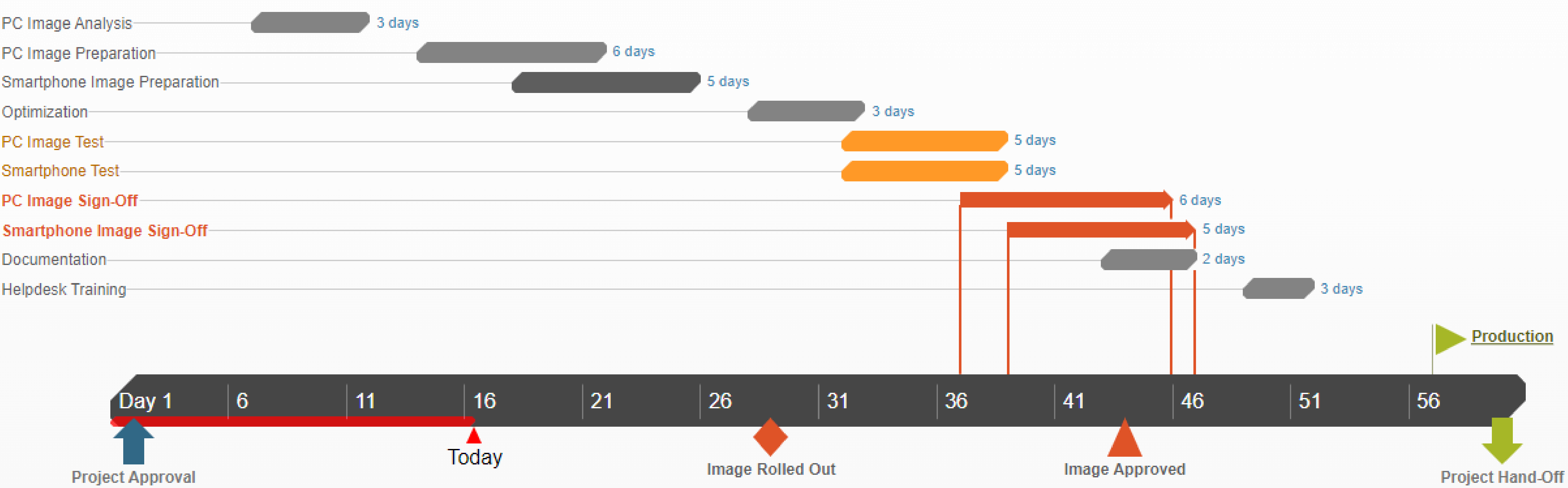
3.5.1 Gantt Charts

Visualizing the Schedule

A Gantt chart is a horizontal bar chart used to illustrate the project schedule.

- **Axis:** Time is on the X-axis; Activities are on the Y-axis.
- **Bars:** Length represents duration.
- **Utility:** Easy for stakeholders to understand progress and overlaps.

Modern tools like **Microsoft Project** automate the creation of Gantt charts from WBS and dependency data.


PC Image Analysis
3 days
PC Image Preparation
6 days
Smartphone Image Preparation
5 days
Optimization
3 days
PC Image Test
5 days
Smartphone Test
5 days
PC Image Sign-Off
6 days
Smartphone Image Sign-Off
5 days
Documentation
2 days
Helpdesk Training
3 days
Production
Day 1
6
11
16
21
26
31
36
41
46
51
56
Project Approval
Today
Image Rolled Out
Image Approved
Project Hand-Off

Topic 3 Summary

Define

Set SMART objectives and decompose work using WBS.

Assign

Clarify roles with the RACI matrix.

Calculate

Use Network Logic (ES, EF, LS, LF) to find the Critical Path.

Visualize

Communicate the plan effectively using Gantt Charts.

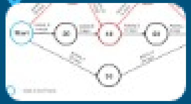
Questions?

Thank you for your attention.

Image Sources

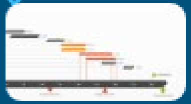
https://itpmschool.com/wp-content/uploads/2023/10/Work-breakdown-structure-chart.webp

Source: itpmschool.com



https://www.project-risk-manager.com/wp-content/uploads/2017/10/PERT-Network-Diagram-Rev0.png

Source: www.project-risk-manager.com



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